

Bronchial Branch Tracing Workflow

Step-by-step reproduction checklist

Aim: CT -> airway segmentation -> CT-verified route -> 3D model -> virtual bronchoscopy -> branch-tracing output.

Important: This is a research and pre-procedural planning workflow. It does not provide real-time bronchoscope localisation.

Before starting

- Install 3D Slicer 5.12.4 or the documented project version.
- Install SlicerTotalSegmentator, SlicerVMTK and the PyTorch/NNUNet dependencies required by TotalSegmentator.
- Place the supplied scripts in an accessible folder: 1 BBT CT Series Selector v1.0; 2 Make 3D visible; 3 BBT Continuous Smooth Route v4.0; 4 BBT Virtual Branch v7.0.
- Use a de-identified thin-section chest CT for the reproduction exercise.

1. Import the CT

- Open 3D Slicer and import the DICOM examination using the DICOM module.
- Load the chest CT and run 1 BBT CT Series Selector v1.0.
- Select the best thin-section reconstruction for airway analysis.
- **CHECK:** Scroll through axial, coronal and sagittal images and confirm that this is the intended CT series.

2. Mark the lesion

- Open Markups and create a fiducial point.
- Place it in the centre of the target lesion and rename it exactly Lesion.
- **CHECK:** Confirm its position in axial, coronal and sagittal planes.

3. Decide whether to crop

- For the first reproduction attempt, preferably use the full CT.
- If cropping is necessary, create an ROI containing the trachea, main and lobar bronchi, all visible airways along the anticipated route, target region and lesion.
- Use Crop Volume to create the working CT.
- **STOP:** A small lesion-only ROI cannot support a continuous trachea-to-target route.

4. Segment the airways

- Open TotalSegmentator and select the full CT or whole-route cropped CT.
- Select the `Lung_vessels` task and run the segmentation.
- Identify and display the lung airway segmentation over the original CT.
- **MANDATORY CT CHECK:** Inspect axial, coronal and sagittal planes for correct lumen following, leakage, false connections, missing branches and distal termination.

5. Correct the segmentation if necessary

- Compare any missing or doubtful area directly with the source CT.
- Correct manually only where an airway is supported by the CT and record the correction.

■ **STOP:** If an airway cannot be confidently identified on CT, do not invent it or bridge the gap merely to obtain continuity.

6. Mark the final airway target

- Follow the visible airway towards the lesion on the CT.
- Identify the most distal CT-supported and connected airway suitable for approaching the lesion.
- Place a markup inside this airway and rename it exactly RouteTarget.
- **Remember:** Lesion is the clinical target; RouteTarget is the endpoint of the CT-supported airway route.

7. Mark the starting point

- Place another markup inside the tracheal lumen and rename it exactly RouteStart.
- Confirm that RouteStart and RouteTarget lie within the same connected airway segmentation.

8. Generate the 3D airway model

- Run 2 Make 3D visible.
- Inspect the airway tree in 3D and identify the lesion, target airway and intended route.
- **CHECK:** A convincing 3D appearance is not sufficient. Return to the CT whenever anatomy is uncertain.

9. Generate the route

- Run 3 BBT Continuous Smooth Route v4.0.
- Verify that the route begins in the trachea, follows the segmented airway tree, takes the expected daughter branch at each bifurcation, does not cross lung tissue and terminates at RouteTarget.
- **STOP:** If a continuous route cannot be generated, investigate disconnected segmentation or endpoint placement. Do not artificially connect components.

10. Generate virtual bronchoscopy

- Run 4 BBT Virtual Branch v7.0.
- Follow the virtual camera from the trachea towards RouteTarget.
- At each bifurcation ask: *If I were holding the bronchoscope here, which opening should I enter next?*
- Compare each decision with the CT and check camera orientation and clock-face direction.
- **Remember:** Virtual bronchoscopy is a CT-derived simulation, not live localisation.

11. Create the branch-tracing sequence

- For every relevant bifurcation, identify the chosen daughter bronchus and capture the virtual bronchoscopic view.
- Record the branch sequence and approximate clock-face direction where useful.
- Number decisions sequentially: B1, B2, B3, and so on.

12. Final verification

- Review the complete path: Trachea -> B1 -> B2 -> B3 -> ... -> final airway -> lesion.
- For every step confirm: the bronchus is visible on CT; it connects to the previous bronchus; the chosen daughter branch is correct; the virtual view is compatible with the CT; uncertain/manual areas are documented; and the route stops at the last CT-supported airway.

13. Save the case

- Save the Slicer scene, airway segmentation, centreline/route, and the Lesion, RouteStart and RouteTarget markups.

- Save representative 3D images, bifurcation images and the virtual bronchoscopy video if required.
- Record the CT series, Slicer and extension versions, script versions, manual corrections, segmentation problems and unresolved uncertainty.

THE RULE TO REMEMBER: THE CT WINS.

If the segmentation, 3D model, centreline or virtual bronchoscopy disagrees with the source CT, the generated output must not be accepted as anatomical truth. The software can extract and present a route; it cannot establish anatomy that is not supported by the CT.